Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2000-I-Q01] | Part A (Q1-Q5)

Answer: 008

1. Any factor containing both primes $2, 5$ ends in zero. Consequently the only possible factorization avoiding zeros is $2^n \cdot 5^n$; a factor 1 would leave the zero-ending number 10^n.

2. For $1 \leq n \leq 7$, neither 2^n nor 5^n has a zero digit. At $n = 8$, $5^8 = 390625$ contains one. Thus the answer is 8.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2002-II-Q05] | Part A (Q1-Q5)

Answer: 042

1. Prime-exponent comparison shows failure precisely when $a < 6n$ or $a < 6m$. If $n, m \geq 1$, then $a \geq 3 \cdot 2^n \geq 6n$ and $a \geq 2 \cdot 3^m \geq 6m$, so there are no failures.

2. For $m = 0$, $2^n < 6n$ holds exactly at $n = 1, 2, 3, 4$; after $n = 5$, induction preserves the reverse inequality. For $n = 0$, $3^m < 6m$ holds only at $m = 1, 2$.

3. The integers are $2, 4, 8, 16, 3, 9$, summing to 42.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q04] | Part A (Q1-Q5)

Answer: 124

1. The five-adic valuation controls trailing zeros. An integer j occurs in $101 - j$ of the factorials.

2. Multiples $5, 10, \ldots, 100$ contribute $96 + 91 + \cdots + 1 = 970$. Multiples $25, 50, 75, 100$ contribute an additional $76 + 51 + 26 + 1 = 154$. No factor 125 occurs.

3. Thus $N = 1124$, remainder 124.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-II-Q03] | Part A (Q1-Q5)

Answer: 049

1. Count odd multiples of $3, 9, 27, 81$ separately, each adding a valuation layer. Their counts are $33, 11, 4, 1$, with no multiple of 243.

2. The exponent is $33 + 11 + 4 + 1 = 49$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2007-I-Q01] | Part A (Q1-Q5)

Answer: 083

1. For u^2 divisible by $2^3 \cdot 3$, the integer u must be divisible by $2^2 \cdot 3 = 12$, and this condition is sufficient.

2. The positive multiples of twelve below one thousand are $12, 24, \ldots, 996$, numbering 83.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2010-II-Q03] | Part A (Q1-Q5)

Answer: 150

1. A difference d occurs $20-d$ times. Count one contribution for each factor of 2, another for each factor of 4, and similarly for $8, 16$.

2. The contributions are $\sum_{j=1}^{9}(20-2j)=90$, $\sum_{j=1}^{4}(20-4j)=40$, $(20-8)+(20-16)=16$, and $20-16=4$.

3. The total valuation is $90+40+16+4=150$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2015-I-Q03] | Part A (Q1-Q5)

Answer: 307

1. Write $16p = n^3 - 1 = (n-1)(n^2+n+1)$. The integer n is odd, making the second factor odd; thus 16 divides $n-1$.

2. Put $n - 1 = 16k$. Then $p = k(n^2+n+1)$, so primality forces $k = 1$, since $n^2+n+1 > 1$. Hence $n = 17$ and $p = 307$.

3. No prime at most $\sqrt{307} < 18$ divides 307: test $2, 3, 5, 7, 11, 13, 17$. Also $16 \cdot 307 + 1 = 17^3$, verifying the answer.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2019-II-Q04] | Part A (Q1-Q5)

Answer: 187

1. Classify non-five faces by exponent parities of two and three: neutral $1, 4$ (two choices), and single classes $2, 3, 6$. A square product requires an even number of fives, and counts of $2, 3, 6$ either all even or all odd.

2. With zero fives, all-even counts contribute $16 + 72 + 18 + 3 = 109$ sequences and all-odd counts contribute $4! \cdot 2 = 48$, total 157. With two fives, choose positions in six ways and the remaining equal parity classes in $2^2 + 1 + 1 + 1 = 7$ ways, total 42. Four fives gives one.

3. Favorable count 200 of 1296, probability $25/162$, so the answer is 187.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2020-II-Q01] | Part A (Q1-Q5)

Answer: 231

1. Factor $20^{20} = 2^{40}5^{20}$. Necessarily $m = 2^a5^b$, and $n = 2^{40-2a}5^{20-2b}$.

2. Nonnegative remaining exponents require $0 \le a \le 20$ and $0 \le b \le 10$. Each exponent pair determines exactly one n, giving $21 \cdot 11 = 231$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2021-I-Q03] | Part A (Q1-Q5)

Answer: 050

1. A positive integral difference must be $2^b(2^c-1)$ with integers $b \geq 0, c \geq 1$. Negative lower exponents would leave an odd numerator over a power of two, so cannot give an integer. Unique odd parts ensure no double counting.

2. For $c = 1, \ldots, 9$, the admissible counts of b are $10, 9, 8, 7, 6, 4, 3, 2, 1$. For $c \geq 10$ even the smallest value exceeds 999.

3. These counts sum to 50.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2023-I-Q04] | Part A (Q1-Q5)

Answer: 012

1. The prime exponents of $13!$ are $(10, 5, 2, 1, 1, 1)$ for primes $(2, 3, 5, 7, 11, 13)$. Each exponent in m must have the same parity and lie between zero and the corresponding exponent.

2. Independent exponent choices factor the sum into $(1 + 4 + \cdots + 4^5)(3 + 27 + 243)(1 + 25) \cdot 7 \cdot 11 \cdot 13$.

3. These factors are $1365 = 3 \cdot 5 \cdot 7 \cdot 13$, $273 = 3 \cdot 7 \cdot 13$, $26 = 2 \cdot 13$, and $1001 = 7 \cdot 11 \cdot 13$. The resulting exponents are $(1, 2, 1, 3, 1, 4)$, summing to 12.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q09] | Part B (Q6-Q10)

Answer: 046

1. The logarithm sum gives $a^{12}r^{66} = 2^{6018}$, hence $a^2r^{11} = 2^{1003}$. Thus $a = 2^u, r = 2^v$ with nonnegative exponents satisfying $2u + 11v = 1003$.

2. v is odd and $1 \leq v \leq 91$, giving forty-six choices. Each determines a nonnegative integer u, so the count is 46.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2009-II-Q07] | Part B (Q6-Q10)

Answer: 401

1. The summand equals $\binom{2i}{i}/4^i$, so its reduced denominator is purely a power of two. Its original numerator is odd and $(2i)!! = 2^i i!$, hence its reduced denominator exponent is $e_i = i + v_2(i!)$.

2. These exponents strictly increase. With common denominator $2^{e_{2009}}$, only the final summand contributes an odd numerator; all others contribute even numerators. Thus no final two-factor cancellation occurs, and $b = 1$.

3. Legendre gives $v_2(2009!) = 1004 + 502 + 251 + 125 + 62 + 31 + 15 + 7 + 3 + 1 = 2001$. Hence $a = 4010$ and $ab/10 = 401$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2016-II-Q08] | Part B (Q6-Q10)

Answer: 728

1. The product N factors as $3^2 \cdot 7 \cdot 11 \cdot 17 \cdot 31 \cdot 41 \cdot 61$. For ordered triples, each of the six exponent-one primes has three destinations; the exponent of3 splits into three nonnegative parts in $\binom{4}{2} = 6$ ways. This gives $6 \cdot 3^6 = 4374$.

2. If two entries equal v, then $v^2 \mid N$, so $v = 1$ or3. The two multisets $(1, 1, N)$ and $(3, 3, N/9)$ each have three orderings. An all-equal triple is impossible.

3. Remove these six ordered triples and divide by six for the distinct-entry permutations: $(4374 - 6)/6 = 728$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2019-I-Q07] | Part B (Q6-Q10)

Answer: 880

1. Exponentiating shows only primes two and five occur. For either prime, let its exponents be a, b. Then $a + 2\min(a, b) = 60$, $b + 2\max(a, b) = 570$.

2. If $a > b$, solving yields negative b, impossible. Thus $a \leq b$, giving $3a = 60, 3b = 570$, or $a = 20, b = 190$.

3. Both primes have these exponents, so $m = 40, n = 380$ and $3m + 2n = 880$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2020-I-Q10] | Part B (Q6-Q10)

Answer: 407

1. Every prime of n also divides m, so it divides $m+n$ and is at least 11. Since $n \nmid m$, some such prime has exponents $a = v_p(n) > b = v_p(m) \geq 1$. Divisibility requires $bm \geq an$.

2. A candidate is $n = 121, m = 286$: $v_{11}(n^n) = 242 \leq 286 = v_{11}(m^m)$ and $407 = 11 \cdot 37$ is coprime to 210.

3. To rule out smaller sums, assume $m + n < 407$. An exponent $a \geq 3$ forces $n \geq 1331$, impossible. Hence $a = 2, b = 1$, so $m \geq 2n$ and $n \leq 135$. Only $n = 121$ can contain a square of a prime at least 11 in this range.

4. For $242 \leq m < 286$, the multiples of 11 are $242, 253, 264, 275$. The first is divisible by 121; the others give sums divisible respectively by $2, 5, 2$. None works. The minimum is 407.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2000-I-Q11] | Part C (Q11-Q15)

Answer: 248

1. Since $1000 = 2^3 5^3$, coprimality lets each prime appear in at most one of a, b. Thus the exponent differences range independently from -3 to 3, each once.

2. Consequently $S = (\sum_{j=-3}^{3} 2^j)(\sum_{j=-3}^{3} 5^j) = (127/8)(19531/125) = 2480437/1000$. Hence $S/10 = 248.0437$, whose floor is 248.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q13] | Part C (Q11-Q15)

Answer: 899

1. For $1 \leq j < n$, exactly 2^{n-1-j} summands have value 2^j, while the endpoint contributes 2^n. Thus $S_n = (n-1)2^{n-1} + 2^n = (n+1)2^{n-1}$.

2. Even n gives odd two-adic valuation and fails. For odd n, the power of two is already square, so $n+1$ must be square. The largest even square at most one thousand is 900, giving $n = 899$.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-II-Q14] | Part C (Q11-Q15)

Answer: 063

1. Among zero-padded strings from 0 to $10^n - 1$, each digit occurs $n10^{n-1}$ times. The final integer 10^n contributes one. Hence $S_n = 1 + n10^{n-1}H_9$, where $H_9 = 7129/2520$ is reduced.

2. As $2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$, integrality requires $63 \mid n$, because powers of ten supply no factors three or seven. The first such $n = 63$ also supplies all factors two and five, so it works and is minimal.

Source / 来源

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2016-I-Q12] | Part C (Q11-Q15)

Answer: 132

1. The expression is odd. Its residue sets modulo $3, 5, 7$ are respectively $\{1, 2\}, \{1, 2, 3\}, \{2, 3, 4, 6\}$, none containing zero. Thus every prime factor is at least 11.

2. The first possible products of at least four such primes are $11^4 = 14641$ and $11^3 \cdot 13 = 17303$. The first would require discriminant 58521, strictly between 241^2 and 242^2, so is impossible.

3. The expression increases for positive integral m, and at $m = 132$ it equals $132 \cdot 131 + 11 = 17303 = 11^3 \cdot 13$. No smaller value can be a qualifying prime product except the excluded 11^4. Hence the minimum is 132.

Source / 来源